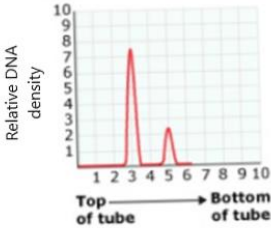
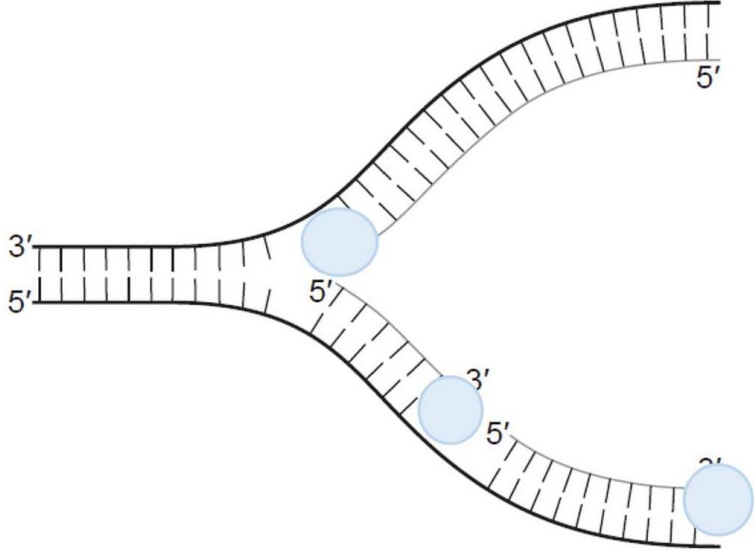


Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3.	(a)	(i)		Start– 2 nd graph Division 1 – 3 rd graph Division 2 – 1 st graph All correct (1)		1		1		
		(ii)		 2 peaks (1) Peaks in correct place (3,5) (1) Peaks relative heights correct 3:1 (1)		3		3	2	

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		One X correct (1) cross should be centred on the newly synthesised strand. 		1		1		
		(ii)		Unwind DNA/ break <u>hydrogen</u> bonds between bases/ {separates strands of / unzips} DNA	1			1		
		(iii)		It shows that {each original strand of DNA/ the original DNA molecule} acts as a template (for a new strand). (1) New (double stranded) DNA contains one old and one new strand (1)	2			2		

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)			Any three (x1) from: A. The {nucleotide/base} sequence (in the DNA) is different / triplet code is different (1) B. This will lead to different mRNA being produced (during transcription)/ change the codons in the mRNA (1) C. And the different amino acid being incorporated into the protein (during translation) / different sequence of amino acids (1) D. The protein may {not function correctly / be non functional}/ different {polypeptide/ protein} formed (1) Reject wrong protein E. due to the changed {tertiary structure/ 3D shape} (1)		3		3		
				Question 3 total	3	8	0	11	2	